CHRIST (Deemed to be University), Bangalore – 560 029

Department of Mathematics

END SEMESTER EXAMINATION – April/May 2024

UG II Semester

Programme Name: Bachelor of Computer Applications **Max. Marks: 100**

Course Name: Discrete Mathematics **Time: 3 hours**

Course Code: BCA104-2

General Instructions

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / Course title & number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place in the hall.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer booklet. No sheets should be detached from the answer booklet.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer booklet.

Course Outcomes (COs): The students will able to

CO1: Demonstrate a working knowledge of set notation and elementary set theory and recognize the connection between set operations and logic.

CO2: Apply the different properties of injections, surjections, bijections, compositions, and inverse functions.

CO3: Understand different types of matrices and operations among matrices.

CO4: Determine when a relation is reflexive, symmetric, antisymmetric, transitive, apply the properties of equivalence relations and partial orderings, and explain the connection between equivalence relations.

CO5: Effective express the elementary concepts and results of Number theory.

SECTION A

Answer ALL the questions **8 × 3 = 24**

Q. No	Questions	CO	RBT
1	Find the symmetric difference of $\{a, c, d\}$ and $\{a, b, c\}$.	1	L1
2	Show that the set of odd positive integers is a countable set. **(OR)** Give an example of a function from N to N that is a) one-to-one but not onto. b) onto but not one-to-one. c) both onto and one-to-one.	2	L2
3	Find the domain, codomain and range of the function that assigns last 2 bits of bit strings of length 2 or greater to that string.	2	L4
4	Define POSET of a set with an example.	2	L3
5	Give an example of an equivalence relation on a set whose equivalence classes are finite.	3	L2

	(OR) How can a directed graph of a relation R on a finite set A be used to determine whether the relation is reflexive, symmetric?		
6	State division algorithm with an example for integers.	4	L1
7	Find the greatest common divisor of $(12, 36, 60, 108)$. **(OR)** Find the least common multiple of $[120, 500]$.	4	L4
8	Find all the incongruent solutions of $16x \equiv 8\ (mod\ 28)$.	4	L3

SECTION B

Answer ALL the questions **7 × 8 = 56**

Q. No	Questions	CO	RBT
9	In a survey of 60 people it was found that, 25 read weekly magazines, 26 read fortnightly magazines, 26 read monthly magazines, 9 read both weekly and monthly magazines,11 read both weekly and fortnightly magazines, 8 read both monthly and fortnightly magazines and 3 read all 3 magazines. Develop the Venn diagram and hence determine the number of people who read a) at least one of the three magazines. b) exactly one magazine.	1	L5
10	Prove that if A and B are countable sets, then $A \cup B$ also countable. **(OR)** If x is a real number, prove that $\lfloor 2x \rfloor = \lfloor x \rfloor + \left\lfloor x + \frac{1}{2} \right\rfloor$.	2	L1
11	Given $f: A \to B$ defined by $f(x) = \frac{x-3}{2x-1}$, $x \neq \frac{1}{2}$. Determine f is invertible if so, write $f^{-1}\left(\frac{2}{3}\right)$.	2	L5
12	Let R_1 and R_2 be relations on a set A represented by the matrices $M_{R_1} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ and $M_{R_2} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$, then find the matrices that represent a) $R_1 \cup R_2$ b) $R_1 \cap R_2$ c) $R_2 \circ R_1$ d) $R_1 \circ R_1$ e) $R_1 \oplus R_2$ **(OR)** Let $R_1, R_2: A \to B$ be the relations on the sets $A = \{1, 2, 3\}$ and $B = \{1,2,3,4\}$ defined by $R_1 = \{(1,1), (2,2), (3,3)\}$ and $R_2 = \{(1,1), (1,2), (1,3), (1,4)\}$. Then find $R_1 \cup R_2$, $R_1 \circ R_2$, $R_2 - R_1$, $R_1 \cap R_2$, $R_1 \oplus R_2$ and its corresponding relation matrices.	3	L3
13	Consider the relation $R = \{(1, 2), (1,3), (1,4), (2,3), (2,4), (3, 4)\}$ on the set $\{1, 2, 3, 4\}$. Find the reflexive, symmetric and transitive closures of R.	3	L4
14	Find the octal expansion of $(12345)_{10}$ and hexadecimal expansions of $(177130)_{10}$.	4	L2
15	State Euclidean algorithm for integers, hence obtain the greatest common divisor of [414, 662]. **(OR)**	4	L4, L5

	Determine the following congruences are solvable or not. (i) $8x \equiv 10\ (mod\ 6)$ (ii) $2x \equiv 3\ (mod\ 4)$ (iii) $4x \equiv 7\ (mod\ 5)$ (iv) $2x \equiv 3\ (mod\ 5)$		

SECTION C

Answer ALL the questions **2 × 10 = 20**

Q. No	Questions	CO	RBT
16	Prove that the set of real numbers are uncountable set. (OR) Draw the Hasse diagram for the poset $(S, \mid)$, where the set S consist $\{2, 4, 6, 9, 12, 18, 27, 36, 48, 60, 72\}$. Hence answer the following a) Find the maximal and minimal elements. b) Is there a greatest and least element? if so write it. c) Find all upper bounds of $\{2, 9\}$ and write the least upper bound of $\{2, 9\}$, if it exists. d) Find all lower bounds of $\{60, 72\}$ and write the greatest lower bound of $\{60, 72\}$, if it exists	1, 2	L3, L4
17	State the Chinese Remainder theorem, and hence solve the following system of linear congruences, $x \equiv 2\ (mod\ 3)$, $x \equiv 3\ (mod\ 5)$, $x \equiv 2\ (mod\ 7)$.	4	L3

Revised Bloom's Taxonomy (RBT) Levels:		
L1 – Remembering	L2 – Understanding	L3 – Applying
L4 – Analyzing	L5 – Evaluating	L6 – Creating